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LaTeX Documentation-Generation System

Desk Instruction DI-LATEX-DOCGEN — v1.0

Documentation-Generation Project ■ April 2026

OPR: Open Source LaTeX Templates

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1. Purpose

This desk instruction describes how to use the LaTeX Documentation-Generation template system to produce professional documents. It covers the available document types, directory conventions, how to create a new document from a template, how to build PDFs, and how to contribute new templates.

NOTE

This document is itself built from the Desk Instruction template. Every callout box, procedure step, and formatting element you see here is available in your own desk instructions via the commands documented in Section 4.

2. Available Document Types

The Documentation-Generation project provides base templates for the following document types:

Type	Directory	Use Case
Attestation	Attestations/	Formal attestation or certification statements
Decision Document	DecisionDocument/	Structured decision records with rationale
Decision Memorandum	DecisionMemorandum/	Short-form memos documenting a decision
Desk Instruction	DeskInstruction/	Procedural guides, how-to documents, training aids
Meeting Agenda	MeetingAgenda/	Structured meeting agendas with action items
Slide Deck	SlideDecks/	Beamer-based presentations
Software ICD	SoftwareICD/	Software Interface Control Documents
Technical Report	TechnicalReport/	Assessment reports, analysis, DRY reviews

Each type follows the same directory layout:

```
DocumentType/
templates/ Base .tex template(s) shared by all instances
examples/ Example documents built from the template
```

3. Creating a New Document

3.1 Step-by-Step Procedure

1. Choose the document type.

Determine which template matches your need from the table above. If none fits, see Section 6 for creating a new template type.

2. Copy the example file.

Navigate to the `examples/` directory of your chosen type and copy the example `.tex` file to your working directory.

CAUTION

Do not edit the base template in `templates/` directly. The template is shared by all documents of that type. Always work from a copy in your own directory or in `examples/`.

3. Edit the variables block.

Open the copied `.tex` file and update the variable definitions at the top. Each template documents its required and optional variables in comments. For example, a Desk Instruction requires:

```
\newcommand{\DITitle}{Your Title Here}
\newcommand{\DIID}{DI-YOUR-ID}
\newcommand{\DIDate}{Month YYYY}
\newcommand{\DIVersion}{v1.0}
\newcommand{\DIAuthor}{Your Name or Org}
\newcommand{\DISubject}{One-line description}
\newcommand{\DIOPR}{Your Office}
```

4. Replace the content commands.

Each template has one or more content commands (`\DIContent`, `\DIReferences`, etc.) that you override with `\renewcommand`. Write your content inside these commands.

5. Verify the `\input` path.

Ensure the `\input{../templates/desk_instruction_base.tex}` path is correct relative to your file's location.

COMMON MISTAKE

A common mistake is copying the example to a different directory depth without updating the relative path to the template. If `pdflatex` says "File not found," check this path first.

6. Build the PDF.

Run `pdflatex` twice (the second pass resolves cross-references and the table of contents):

```
pdflatex YourDocument.tex
pdflatex YourDocument.tex
```

Or if a Makefile is available: `make`

7. Review the output.

Open the generated PDF and verify: title page variables, table of contents, section numbering, and callout-box rendering.

4. Custom Commands Reference

The Desk Instruction template provides the following custom commands. All are optional — use them when they add clarity.

4.1 Callout Boxes

NOTE

Use `\dinote{text}` for informational notes that provide helpful context.

CAUTION

Use `\dicaution{text}` for cautions about non-obvious constraints or pitfalls.

WARNING

Use `\diwarning{text}` for warnings about actions that could cause data loss or process violations.

WHAT TO SAY

Use `\discript{text}` for “what to say” scripts — exact wording an operator should use in a meeting or conversation.

COMMON MISTAKE

Use `\digotcha{text}` for common mistakes that experienced practitioners have learned to avoid.

4.2 Inline Commands

- `\distep{title}` — use inside an `enumerate` environment to create a bold-titled procedure step.
Example: `\distep{Open the file.}`
- `\diref{doc}{paragraph}{description}` — inline reference callout. Renders as:
SSP 57000 1.3.5 — Ethernet Interface Requirements

4.3 Section Structure

The template provides automatic table of contents, navy-colored section headers with teal accent rules, and three levels of headings (`\section`, `\subsection`, `\subsubsection`).

Optional blocks are emitted only if defined:

- `\DIReferences` — rendered as a “References” section
- `\DIGlossary` — rendered as a “Glossary” section
- `\DIAppendix` — rendered after `\appendix` command (sections become A, B, C...)

5. Design Conventions

5.1 Variable Naming

All template variables use a two-letter prefix matching the document type:

Type	Prefix
Desk Instruction	<code>\DI...</code>
Technical Report	<code>\Report...</code>
Decision Memo	<code>\DM...</code> or direct names

5.2 Color Palette

The Desk Instruction template uses a consistent color palette:

Name	Hex	Use
diNavy	#0B2545	Headings, primary text emphasis
diAccent	#1A8FA6	Title-page rules, accent elements
diMuted	#6C757D	Footer text, secondary metadata
diBgLight	#F8F9FB	Light background tint

Other document types may define their own palettes.

5.3 File Naming

- Templates: `<type>_base.tex` in `templates/`
- Examples: `DI-<ShortName>.tex` in `examples/`
- Built PDFs: same base name with `.pdf` extension

6. Creating a New Template Type

If none of the existing templates fits your document type:

1. Create the directory structure.

```
mkdir -p NewType/templates NewType/examples
```

2. Write the base template.

Copy the closest existing template (e.g., `desk_instruction_base.tex`) and adapt:

- Rename the variable prefix (e.g., `\NT...`)
- Adjust the header/footer to reflect the new document type name
- Add or remove custom commands as needed

- Update the `\hypersetup` metadata

3. Write the example document.

The example should be **self-documenting** — it explains how to use the template *by being an instance of it*. This is the convention across the project.

CAUTION

The example must not contain organization-specific, proprietary, or classified content. This is a public repository. Use generic or meta-referencing examples.

4. Build and verify.

Build the example, verify TOC, headers, footers, callout boxes, and page numbering.

5. Update the project documentation.

Add the new type to the table in `AGENTS.md` (if it exists) and to this desk instruction's "Available Document Types" table.

7. Troubleshooting

7.1 pdflatex fails with "File not found"

Check the `\input{...}` path relative to your `.tex` file's location. Use `../templates/` when building from `examples/`.

7.2 Table of contents is empty

Run `pdflatex` twice. The first pass generates the `.toc` file; the second pass reads it.

7.3 tcolorbox package not found

The callout boxes require the `tcolorbox` package. Install it via your TeX distribution's package manager (`tlmgr install tcolorbox` for TeX Live).

7.4 Fonts look wrong

The template uses `lmodern` (Latin Modern) with sans-serif default. Ensure `lmodern` is installed (`tlmgr install lm`).

References

- **LaTeX Documentation-Generation** — GitHub repository containing all templates.
- **The L^AT_EX Companion** (3rd ed.) — Mittelbach et al. General L^AT_EX reference.
- **tcolorbox documentation** — <https://ctan.org/pkg/tcolorbox>. Reference for callout-box styling.
- **fancyhdr documentation** — <https://ctan.org/pkg/fancyhdr>. Header/footer configuration.
- **booktabs documentation** — <https://ctan.org/pkg/booktabs>. Table formatting.

Glossary

Term	Definition
DI	Desk Instruction
DM	Decision Memorandum
OPR	Office of Primary Responsibility
PDF	Portable Document Format
TOC	Table of Contents
TR	Technical Report
ICD	Interface Control Document
CEF	Change Evaluation Form